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ANTENNA STRUCTURE AND ELECTRONIC DEVICE INCLUDING SAME

Abstract

An example electronic device may include: a first housing including a first conductive portion, a first non-conductive portion, and a first segmented portion extending from the first conductive portion, a second housing configured to accommodate at least a portion of the first housing and to guide a slide movement of the first housing, the second housing including a second conductive portion, and a flexible display including a first area connected to the first housing and a second area extending from the first area and configured to be bendable or rollable, wherein, from a slide-in state to a slide-out state of the first housing with respect to the second housing, the first conductive portion and the second conductive portion may be spaced apart from each other, and in the slide-in state of the first housing with respect to the second housing, at least a portion of the first non-conductive portion may overlap the second conductive portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/866,101, filed Jul. 15, 2022, which is a continuation of International Application No. PCT/KR2022/009182 designating the United States, filed on Jun. 28, 2022, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2021-0094997, filed on Jul. 20, 2021, in the Korean Intellectual Property Receiving Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND
Field
[0002] The disclosure relates to an antenna structure and an electronic device including the same.
Description of Related Art
[0003] With the development of electronic, information, and communication technologies, various functions are being integrated into one portable communication device or electronic device. For example, a smartphone includes functions of a sound reproduction device, an imaging device, and a digital diary, in addition to a communication function, and further various functions may be implemented in the smartphone through additional installation of applications.
[0004] As the use of personal or portable communication devices, such as smartphones, has become common, users' demands for portability and ease of use are increasing. For example, a touch screen display may provide a virtual keypad that replaces a mechanical input device (e.g., a button input device) while serving as an output device that outputs a screen (e.g., visual information). Accordingly, a portable communication device or an electronic device is capable of providing the same or further improved usability (e.g., a larger screen) while being miniaturized. On the other hand, with the commercialization of flexible, for example, foldable or rollable displays, the portability and ease of use of electronic devices are expected to further improve.
[0005] In an electronic device including a flexible display expandable by sliding, structures of the electronic device may move (e.g., slide, rotate, or pivot) relative to each other. In this case, some structures (e.g., the first housing and a partial area of the flexible display) may move into or away from another structure (e.g., the second housing), and a radiator portion providing an antenna function of the some structures (e.g., the first housing) and a portion (e.g., the conductive portion) of the other structure (e.g., the second housing) may overlap each other. Coupling between the antenna radiator and the portion (e.g., a metal portion) and/or a change in the coupling area of the antenna radiator due to a relative movement may generate unnecessary capacitance, thereby deteriorating antenna performance.
SUMMARY
[0006] Embodiments of the disclosure provide an antenna structure that is capable of providing stable radiation performance by including a segmented portion disposed between an antenna radiator included in a first housing of an electronic device and a portion (e.g., a conductive portion)

of a second housing overlapping the antenna radiator.
[0007] An electronic device according to various embodiments of the disclosure may include: a first housing including a first conductive portion, a first non-conductive portion, and a first segmented portion extending from the first conductive portion, a second housing configured to accommodate at least a portion of the first housing and to guide a slide movement of the first housing, the second housing including a second conductive portion, and a flexible display including a first area connected to the first housing and a second area extending from the first area and configured to be bendable or rollable, wherein, from a slide-in state to a slide-out state of the first housing with respect to the second housing, the first conductive portion and the second conductive portion may be spaced apart from each other, and in the slide-in state of the first housing with respect to the second housing, at least a portion of the first non-conductive portion may overlap the second conductive portion.
[0008] An electronic device according to various example embodiments of the disclosure may include: a first housing including a first conductive portion, a first non-conductive portion, and a first segmented portion extending from the first conductive portion, a second housing configured to accommodate at least a portion of the first housing and to guide a slide movement of the first housing, the second housing including a second conductive portion, and a flexible display including a first area connected to the first housing and a second area extending from the first area and configured to be bendable or rollable, wherein, from a slide-in state to a slide-out state of the first housing with respect to the second housing, the first conductive portion and the second conductive portion may be spaced apart from each other with the first segmented portion interposed therebetween.
[0009] In an electronic device according to various embodiments, the relative movements of housings can be stably performed.
[0010] In an electronic device according to various embodiments, it is possible to provide an antenna structure that provides stable radiation performance of an antenna radiator designed in a first housing (or a second housing) when the first housing and the second housing perform a relative movement operation.
[0011] In an electronic device according to various embodiments, it is possible to provide an antenna structure that is capable of maintaining a resonance characteristic similarly between slide-in and slide-out operations of a first housing and a flexible display with respect to a second housing.
[0012] In an electronic device according to various embodiments, a slidable segmented portion is provided between an antenna radiator of a first housing and a second housing so that interference of the antenna radiator can be prevented and/or reduced when the first housing is moved relative to the second housing.
[0013] Effects that can be obtained in the disclosure are not limited to those described above, and other effects not described above will be clearly understood by a person skilled in the art to which the disclosure belongs based on the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0014] The above and other aspects, features and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:
[0015] FIG. **1** is a block diagram illustrating an example electronic device in a network environment according to various embodiments;
[0016] FIG. **2** is a diagram illustrating a state in which a second display area of a flexible display is

accommodated in a second housing, according to various embodiments;
[0017] FIG. **3** is a diagram illustrating a state in which the second display area of the flexible display is exposed to the outside of the second housing, according to various embodiments;
[0018] FIG. **4** is an exploded perspective view illustrating the electronic device according to various embodiments;
[0019] FIG. **5**A is a perspective view of a first housing and a second housing illustrating a segmented portion in the closed state (the slide-in state) of the electronic device, according to various embodiments;
[0020] FIG. **5**B is a perspective view of the first housing and the second housing illustrating the segmented portion in the opened state (the slide-out state) of the electronic device, according to various embodiments;
[0021] FIG. **6** is a perspective view of a side frame of the first housing and a side frame of the second housing illustrating segmented portions, according to various embodiments;
[0022] FIG. **7**A is an enlarged perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is closed according to various embodiments;
[0023] FIG. **7**B is an enlarged perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is open according to various embodiments;
[0024] FIG. **8**A is a diagram illustrating an internal antenna structure of the electronic device in a state in which the electronic device is closed according to various embodiments;
[0025] FIG. **8**B is a diagram illustrating an internal antenna structure of the electronic device in a state in which the electronic device is open according to various embodiments;
[0026] FIG. **9** is a graph showing antenna performance related to the first antenna structure by the sliding operation between the housings in FIGS. **8**A and **8**B according to various embodiments;
[0027] FIG. **10** is a graph showing antenna performance related to the second antenna structure by the sliding operation between the housings in FIGS. **8**A and **8**B according to various embodiments;
[0028] FIG. **11**A is an enlarged perspective view of an area of the first housing and the second housing illustrating a segmented portion in a state in which the electronic device is closed according to various embodiments;
[0029] FIG. **11**B is an enlarged perspective view of an area of the first housing and the second housing illustrating a segmented portion in a state in which the electronic device is open according to various embodiments;
[0030] FIG. **12**A is an enlarged perspective view of an area of the first housing and the second housing illustrating a segmented portion in a state in which the electronic device is closed according to various embodiments;
[0031] FIG. **12**B is an enlarged perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is open according to various embodiments;
[0032] FIG. **13**A is a perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is closed according to various embodiments;
[0033] FIG. **13**B is a perspective view of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is open according to various embodiments;
[0034] FIG. **14**A is a perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is closed according to various embodiments; and
[0035] FIG. **14**B is a perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is open according to

various embodiments.

DETAILED DESCRIPTION

[0036] FIG. **1** is a block diagram illustrating an example electronic device in a network environment according to various embodiments of the disclosure.

[0037] Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In various embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In various embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0038] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0039] The auxiliary processor **123** may control, for example, at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active (e.g., executing an application) state. According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0040] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0041] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0042] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0043] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0044] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0045] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or an external electronic device (e.g., an electronic device **102** (e.g., a speaker or a headphone)) directly or wirelessly coupled with the electronic device **101**.

[0046] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0047] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0048] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

[0049] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0050] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0051] The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0052] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0053] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify or authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0054] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0055] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0056] According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0057] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0058] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the external electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In an embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0059] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0060] It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B, or C”, “at least one of A, B, and C”, and “at least one of A, B, or C”, may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd”, or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with”, “coupled to”, “connected with”, or “connected to” another element (e.g., a second element), the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0061] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic”, “logic block”, “part”, or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0062] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components or operations may be omitted, or one or more other components or operations may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0063] FIG. **2** is a diagram illustrating a state in which a second display area of a flexible display is accommodated in a second housing, according to various embodiments. FIG. **3** is a diagram illustrating a state in which the second display area of the flexible display is exposed to the outside of the second housing, according to various embodiments.

[0064] The state illustrated in FIG. **2** may be understood as the state in which the first housing **201** is closed relative to the second housing **202**, and the state illustrated in FIG. **3** may be understood as the state in which the first housing **201** is opened relative to the second housing **202**. According to an embodiment, the “closed state” or the “opened state” may be understood as the state in which the electronic device is closed or the state in which the electronic device is opened. According to an embodiment, a “slide-in state” or a “slide-out state” may be understood as the state in which the electronic device is closed or the state in which the electronic device is opened.

[0065] Referring to FIGS. **2** and **3**, an electronic device **101** may include housings **201** and **202**. The housings **201** and **202** may include a second housing **202** and a first housing **201** disposed to be movable relative to the second housing **202**. In some embodiments, the electronic device **101** may be interpreted as a structure in which the second housing **202** is disposed to be slidable on the first housing **201**. According to an embodiment, the first housing **201** may be disposed to be reciprocable by a predetermined distance in the illustrated direction (e.g., the direction indicated by arrow {circle around (**1**)}) relative to the second housing **202**. The configuration of the electronic device **101** of FIGS. **2** and **3** may be wholly or partly the same as or similar to that of the electronic device **101** of FIG. **1**.

[0066] According to various embodiments, the first housing **201** may be referred to as, for example, a first structure, a slide unit, or a slide housing, and may be disposed to be reciprocable on the second housing **202**. According to an embodiment, the first housing **201** may accommodate various electrical and electronic components such as a main circuit board and a battery. The second housing **202** may be referred to as, for example, a second structure, a main unit, or a main housing, and may guide the movement of the first housing **101**. A portion of the display **203** (e.g., a first display area A**1**) may be seated on the first housing **201**. According to an embodiment, when the first housing **201** moves (e.g., slides) relative to the second housing **202**, another portion of the display **203** (e.g., a second display area A**2**) may be accommodated inside the second housing **202** (e.g., a slide-in operation) or exposed to the outside of the second housing **202** (e.g., a slide-out operation).

[0067] According to various embodiments, the first housing **201** may include a first plate **211** (e.g., a slide plate). The first plate **211** may include a first surface F**1** providing at least a portion of the first plate **211** and a second surface F**2** facing away from the first surface F**1**. According to an embodiment, the first plate **211** may support at least a portion of the display **203** (e.g., the first display area A**1**). According to an embodiment, the first housing **201** may include a first plate **211**, a (1-1).sup.th sidewall **211***a* extending from the first plate **211**, a (1-2).sup.th sidewall **211***b* extending from the (1-1).sup.th sidewall **211***a* and the first plate **211**, and a (1-3).sup.th sidewall **211***c* extending from the (1-1).sup.th sidewall **211***a* and the first plate **211** and substantially parallel to the (1-2).sup.th sidewall **211***b*.

[0068] According to various embodiments, the second housing **202** may include a second plate (e.g., the second plate **221** and the main case in FIG. **4**), a (2-1).sup.th sidewall **221***a* extending from the second plate **221**, a (2-2).sup.th sidewall **221***b* extending from the (2-1).sup.th sidewall **221***a* and the second plate **221**, and a (2-3).sup.th sidewall **221***c* extending from the (2-1) sidewall **221***a* and the second plate **221** and substantially parallel to the (2-2).sup.th sidewall **221***b*. According to an embodiment, the (2-2).sup.th sidewall **221***b* and the (2-3).sup.th sidewall **221***c* may be configured to be substantially perpendicular to the (2-1).sup.th sidewall **221***a*. According to an embodiment, the second plate **221**, the (2-1).sup.th sidewall **221***a*, the (2-2).sup.th sidewall **221***b*, and the (2-3).sup.th sidewall **221***c* may have a shape opened on one side (e.g., the front surface) to accommodate (or surround) at least a portion of the first housing **201**. For example, the first housing **201** may be coupled to the second housing **202** in a state of being at least partially surrounded and may be slidable in a direction parallel to the first surface F**1** or the second surface F**2** (e.g., the direction indicated by arrow {circle around (**1**)}) while being guided by the second housing **202**. According to an embodiment, the second plate **221**, the (2-1).sup.th sidewall **221***a*, the (2-2).sup.th sidewall **221***b*, and/or the (2-3).sup.th sidewall **221***c* may be integrally configured. According to an embodiment, the second plate **221**, the (2-1).sup.th sidewall **221***a*, the (2-2).sup.th sidewall **221***b*, and/or the (2-3).sup.th sidewall **221***c* may be configured separately and coupled or assembled to each other.

[0069] According to various embodiments, the second plate **221** and/or the (2-1).sup.th sidewall **221***a* may cover at least a portion of the flexible display **203**. For example, at least a portion of the flexible display **203** may be accommodated in the inside of the second housing **202**, and the second plate **221** and/or the (2-1).sup.th sidewall **221***a* may cover a portion of the flexible display **203** accommodated in the inside of the second housing **202**.

[0070] According to various embodiments, the first hosing **201** may be movable in a first direction (e.g., direction {circle around (**1**)}) substantially parallel to the (2-2).sup.th sidewall **221***b* or the (2-3).sup.th sidewall **221***c* to the opened state or the closed state relative to the second housing **202**, and the first housing **201** may be movable to be positioned at a first distance from the (2-1).sup.th sidewall **221***a* in the closed state and at a second distance, which is greater than the first distance, from the (2-1).sup.th sidewall **221***a* in the opened state.

[0071] According to various embodiments, the electronic device **101** may include a display **203**, key input devices **241**, a connector hole **243**, audio modules **247***a* and **247***b*, or camera modules **249***a* and **249***b*. Although not illustrated, the electronic device **101** may further include an indicator (e.g., an LED device) or various sensor modules. The configurations of the display **203**, the audio modules **247***a* and **247***b*, and the camera modules **249***a* and **249***b* of FIGS. **2** and **3** may be wholly or partly the same as or similar to those of the display module **160**, the audio module **170**, and the camera module **180** of FIG. **1**.
[0072] According to various embodiments, the display **203** may include a first display area A**1** and a second display area A**2**. According to an embodiment, the first display area A**1** may be disposed on the first housing **201**. For example, the first display area A**1** may extend substantially across at least a portion of the first surface F**1** and may be disposed on the first surface F**1**. The second display area A**2** may extend from the first display area A**1** and may be inserted into or accommodated in the inside of the second housing **202** (e.g., a structure) according to the sliding of the first housing **201**, or may be exposed to the outside of the second housing **202**.
[0073] According to various embodiments, the second display area A**2** may move while being substantially guided by a roller (e.g., the curved surface **250** in FIG. **4**) mounted on the first housing **201** to be accommodated inside the second housing **202** or in a space defined between the first housing **201** and the second housing **202** or to be exposed to the outside. According to an embodiment, the second display area A**2** may be moved based on the slide movement of the first housing **201** in the first direction (e.g., the direction indicated by arrow {circle around (**1**)}). For example, while the first housing **201** slides, a portion of the second display area A**2** may be deformed into a curved shape at a position corresponding to the curved surface **250** of the first housing **201**.
[0074] According to various embodiments, when viewed from above the first plate **211** (e.g., the slide plate), if the first housing **201** moves from the closed state to the opened state, the second display area A**2** may define a substantially flat surface with the first display area A**1** while being gradually exposed to the outside of the second housing **202**. The display **203** may be coupled to or disposed adjacent to a touch detection circuit, a pressure sensor capable of measuring touch intensity (pressure), and/or a digitizer configured to detect a magnetic-field-type stylus pen. In an embodiment, the second display area A**2** may be at least partially accommodated inside the second housing **202**, and even in the state (e.g., the closed state) illustrated in FIG. **2**, a portion of the second display region A**2** may be visually exposed to the outside. According to an embodiment, irrespective of the closed state or the opened state, a portion of the exposed second display area A**2** may be positioned on a portion (e.g., the curved surface **250** in FIG. **4**) of the first housing, and at a position corresponding to the curved surface **250**, a portion of the second display area A**2** may maintain the curved shape.
[0075] According to various embodiments, the key input devices **241** may be positioned in an area of the first housing **201**. Depending on the external appearance and use state, the electronic device **101** may be designed such that the illustrated key input devices **241** are omitted or an additional key input device(s) is (are) included. According to an embodiment, the electronic device **101** may include a key input device (not illustrated), such as a home key button or a touch pad disposed around the home key button. According to an embodiment, at least some of the key input devices **241** may be disposed on the (2-1).sup.th sidewall **221***a*, the (2-2).sup.th sidewall **221***b*, or the (2-3).sup.th sidewall **221***c* of the second housing **202**.
[0076] According to various embodiments, the connector hole **243** may be omitted according to an embodiment, and may accommodate a connector (e.g., a USB connector) for transmitting and receiving power and/or data to and from an external electronic device. Although not illustrated, the electronic device **101** may include a plurality of connector holes **243**, and some of the connector holes **243** may function as connector holes for transmitting and receiving audio signals to and from an external electronic device. In the illustrated embodiment, the connector hole **243** is disposed in

the (2-3).sup.th sidewall **221***c*, but the disclosure is not limited thereto. The connector hole **243** or another connector hole (not illustrated) may be disposed in the (2-1).sup.th sidewall **221***a* or the (2-2).sup.th sidewall **221***b*.

[0077] According to various embodiments, the audio modules **247***a* and **247***b* may include at least one speaker hole **247***a* or at least one microphone hole **247***b*. One speaker hole **247***a* may be provided as a receiver hole for a voice call, and another one may be provided as an external speaker hole. The electronic device **101** may include a microphone configured to acquire sound, and the microphone may acquire sound outside the electronic device **101** through the microphone hole **247***b*. According to an embodiment, the electronic device **101** may include a plurality of microphones to detect the direction of sound. According to an embodiment, the electronic device **101** may include an audio module in which the speaker hole **247***a* and the microphone hole **247***b* are implemented as a single hole, or a speaker in which the speaker hole **247***a* is excluded (e.g., a piczo speaker).

[0078] According to various embodiments, the camera modules **249***a* and **249***b* may include a first camera module **249***a* and a second camera module **249***b*. The second camera module **249***b* may be positioned in the first housing **201** and may image a subject in a direction opposite to the first display area A**1** of the display **203**. The electronic device **101** may include a plurality of camera modules **249***a* and **249***b*. For example, the electronic device **101** may include at least one of a wide-angle camera, a telephoto camera, and a close-up camera. According to an embodiment, the electronic device **101** may include an infrared projector and/or an infrared receiver to measure the distance to the subject. The camera modules **249***a* and **249***b* may include one or more lenses, image sensors, and/or image signal processors. The first camera module **249***a* may be disposed to face the same direction as the display **203**. For example, the first camera module **249***a* may be disposed around the first display area A**1** or in an area overlapping the display **203**, and when disposed in the area overlapping the display **203**, the first camera module **249***a* may image a subject through the display **203**.

[0079] According to various embodiments, an indicator (not illustrated) of the electronic device **101** may be disposed on the first housing **201** or the second housing **202**, and may provide state information of the electronic device **101** as a visual signal by including a light-emitting diode. A sensor module (not illustrated) of the electronic device **101** may generate an electrical signal or a data value corresponding to an internal operating state of the electronic device **101** or an external environmental state. The sensor module may include, for example, a proximity sensor, a fingerprint sensor, or a biometric sensor (e.g., an iris/face recognition sensor or an HRM sensor). In an embodiment, the sensor module may further include at least one of, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0080] FIG. **4** is an exploded perspective view illustrating the electronic device according to various embodiments.

[0081] Referring to FIG. **4**, the electronic device **101** may include a first housing **201**, a second housing **202**, and a display **203** (e.g., a flexible display, a foldable display, or a rollable display), and an articulated hinge structure **213**. A portion of the display **203** (e.g., the second display area A**2**) may be accommodated in the inside of the electronic device **101** along the curved surface **250** of the first housing **201**.

[0082] The configurations of the first housing **201**, the second housing **202**, and the display **203** of FIG. **4** may be wholly or partly the same as or similar to those of the first housing **201**, the second housing **202**, and the display **203** of FIGS. **2** and **3**.

[0083] According to various embodiments, the first housing **201** may include a first plate **211** and a slide cover **212**. The first plate **211** and the slide cover **212** may be mounted on (e.g., at least partially connected to) the second housing **202**, and may lincarly reciprocate in a direction (e.g., the direction of the arrow {circle around (**1**)} in FIG. **1**) while being guided by the second housing **202**. According to an embodiment, the first plate **211** may include a first surface F**1**, and the first display region A**1** of the display **203** may be substantially mounted on the first surface F**1** to be maintained in the form of a flat plate. The slide cover **212** may protect the display **203** positioned on the first plate **211**. For example, at least a portion of the display **203** may be positioned between the first plate **211** and the slide cover **212**. According to an embodiment, the first plate **211** may be made of a metal material and/or a non-metal (e.g., a polymer) material. According to an embodiment, the first plate **211** may accommodate components of the electronic device **101** (e.g., the battery **289** (e.g., the battery **189** in FIG. **1**) or the circuit board **204**).

[0084] According to various embodiments, the articulated hinge structure **213** may be connected to the first housing **201**. For example, the articulated hinge structure **213** may be positioned between the first plate **211** and the slide cover **212**. According to an embodiment, when the first housing **201** slides, the articulated hinge structure **213** is movable relative to the second housing **202**. In the closed state (e.g., FIG. **2**), substantially most of the structure of the articulated hinge structure **213** may be accommodated in the second housing **202**. According to an embodiment, at least a portion of the articulated hinge structure **213** may move to correspond to the curved surface **250** positioned at an edge of the first housing **201**.

[0085] According to various embodiments, the articulated hinge structure **213** may include a plurality of bars or rods **214**. The plurality of rods **214** may linearly extend to be disposed substantially parallel to the rotation axis R of the roller **250**, and may be arranged in a direction substantially perpendicular to the rotation axis R (e.g., the direction in which the first housing **201** slides).

[0086] According to various embodiments, each rod **214** may rotate while maintaining a parallel state with another adjacent rod **214**. For example, one of the rods **214** may move while rotating about another adjacent rod around at least a portion of the circumference of the same. According to an embodiment, as the first housing **201** slides, the plurality of rods **214** may be arranged to define a curved shape or may be arranged to define a flat shape. For example, as the first housing **201** slides, a portion of the articulated hinge structure **213** facing the curved surface **250** may define a curved surface, and another portion of the articulated hinge structure **213** that does not face the curved surface **250** may define a flat surface. According to an embodiment, the second display region A**2** of the display **203** may be mounted or supported on the articulated hinge structure **213**, and in the open state (e.g., FIG. **3**), at least a portion of the second display region A**2** may be exposed to the outside of the second housing **202** together with the first display region A**1**. In the state in which the second display area A**2** is exposed to the outside of the second housing **202**, the articulated hinge structure **213** may support or maintain the second display area A**2** in the flat state by defining a substantially flat surface. According to an embodiment, the articulated hinge structure **213** may be replaced with a bendable integrated support member (not illustrated).

[0087] According to various embodiments, the second housing **202** may include a second plate **221**, a second plate cover **222**, and/or a third plate **223**. The second plate **221** may support, for example, the electronic device **101** as a whole. The second plate **221** may include one surface on which the first plate **211** is disposed and the other surface on which the printed circuit board **204** is placed. According to an embodiment, the second plate **221** may accommodate components of the electronic device **101** (e.g., the battery **289** (e.g., the battery **189** in FIG. **1**) or the circuit board **204**). The second plate cover **222** may protect various components positioned on the second plate **221**.

[0088] According to various embodiments, on the circuit board **204**, a processor, a memory, and/or an interface may be disposed. The processor may include, for example, one or more of a central processing unit, an application processor, a graphics processing unit, an image signal processor, a sensor hub processor, or a communication processor. According to various embodiments, the circuit board **204** may include a flexible printed circuit board type radio frequency cable (FRC). For example, the circuit board **204** may be disposed on at least a portion of the second plate **221**, and may be electrically connected to an antenna module (e.g., the antenna module **197** in FIG. **1**) and a communication module (e.g., the communication module **190** in FIG. **1**).

[0089] According to an embodiment, the memory may include, for example, a volatile memory or a nonvolatile memory.

[0090] According to an embodiment, the interface may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, an SD card interface, and/or an audio interface. The interface may electrically or physically connect, for example, the electronic device **101** to an external electronic device and may include a USB connector, an SD card/an MMC connector, or an audio connector.

[0091] According to various embodiments, the battery **289** is a device for supplying power to at least one component of the electronic device **101** and may include, for example, a non-rechargeable primary battery, a rechargeable secondary battery, or a fuel cell. At least a portion of the battery **289** may be disposed on substantially the same plane as, for example, the circuit board **204**. The battery **289** may be integrally disposed inside the electronic device **101**, or may be disposed to be detachable from the electronic device **101**.

[0092] According to an embodiment, the third plate **223** may substantially define at least a portion of the exterior of the second housing **202** or the electronic device **101**. For example, the third plate **223** may be coupled to the outer surface of the second plate cover **222**. According to an embodiment, the third plate **223** may be configured integrally with the second plate cover **222**. According to an embodiment, the third plate **223** may provide a decorative effect on the exterior of the electronic device **101**. The second plate **221** and the second plate cover **222** may be manufactured using at least one of a metal or a polymer, and the third plate **223** may be manufactured using at least one of metal, glass, synthetic resin, or ceramic. According to an embodiment, the second plate **221**, the second plate cover **222**, and/or the third plate **223** may be at least partially (e.g., in an auxiliary display area) made of a material that transmits light. For example, in the state in which a portion of the display **203** (e.g., the second display area A**2**) is accommodated in the inside of the electronic device **101**, the electronic device **101** may output visual information using the second display area A**2**. The auxiliary display area may be a portion of the second plate **221**, the second plate cover **222**, and/or the third plate **223** in which the display **203** accommodated in the inside of the second housing **202** is positioned.

[0093] FIG. **5**A is a perspective view of a first housing and a second housing illustrating a segmented portion in the closed state (the slide-in state) of the electronic device, according to various embodiments.

[0094] FIG. **5**B is a perspective view of the first housing and the second housing illustrating the segmented portion in the opened state (the slide-out state) of the electronic device, according to various embodiments.

[0095] FIG. **6** is a perspective view of a side frame of the first housing and a side frame of the second housing illustrating segmented portions, according to various embodiments.

[0096] According to various embodiments, the electronic device **101** may include a first housing **301**, a second housing **302**, and a flexible display **303**. The electronic device **101** may further include an antenna structure, wherein at least one component (e.g., a conductive portion) of the antenna structure may be located in an area of the first housing **201** and/or the second housing **202**. The electronic device **101** may further include various components (e.g., a printed circuit board, a camera module, and a battery) disposed inside the first housing **301** and/or the second housing **302**.

[0097] The configurations of the first housing **301**, the second housing **302**, and the flexible display **303** of FIGS. **5**A, **5**B and **6** may be wholly or partly the same as or similar to those of the first housing **201**, the second housing **202**, and the flexible display **203** of FIGS. **2**, **3** and **4**.

[0098] According to various embodiments, as the first housing **301** (and the flexible display **303** connected to the first housing **301**) slides in or out relative to the second housing **302**, the

electronic device **101** may be in the closed state or the opened state.
[0099] According to various embodiments, the first housing **301** is slidable in a first direction (e.g., the direction {circle around (**1**)}) relative to the second housing **302**. The first housing **301** may include a first conductive portion **410**. For example, the first conductive portion **410** may be a frame (hereinafter, referred to as a "first frame **310**") formed of a metal material of the first housing **201**, and may be used as an antenna radiator. In an embodiment, one end of the first conductive portion **410** may be segmented (e.g., a segmented portion **461**) such that the first conductive portion **410** is used as an antenna radiator.
[0100] According to various embodiments, the second housing **302** may accommodate at least a portion of the first housing **301** and guide the sliding movement of the first housing **301** (and the flexible display **303**). The second housing **302** may include a second conductive portion **420**. For example, the second conductive portion **420** may be a frame (hereinafter, referred to as a "second frame **320**") formed of a metal material of the second housing **202**, and may be used as an antenna radiator. In an embodiment, one end of the first conductive portion **410** may be segmented (e.g., a segmented portion **461**) such that the second conductive portion **420** is used as an antenna radiator. For example, the segmented portion **461** may be located between the first conductive portion **410** and the second conductive portion **420**.
[0101] According to various embodiments, the first frame **310** of the first housing **301** may provide at least a portion of the side surface of the first housing **301**. The first frame **310** may include a (1-1).sup.th side portion **311** (e.g., the (1-2).sup.th sidewall **211***b* in FIG. **3**), a (1-2).sup.th side portion **312** (e.g., the (1-1).sup.th sidewall **211***a* in FIG. **3**) extending from the (1-1).sup.th side portion **311** and disposed in a direction (e.g., the vertical direction) different from that of the (1-1).sup.th side portion **311**, and a (1-3).sup.th side portion **313** (e.g., the (1-3).sup.th sidewall **211***c* in FIG. **3**) extending from the (1-2).sup.th side portion **312** and disposed in the same direction as the (1-1).sup.th side portion **311**. The second frame **320** of the second housing **202** may provide at least a portion of the side surface of the second housing **202**. The second frame **320** may include a (2-1).sup.th side portion **321** (e.g., the (2-2).sup.th sidewall **221***b* in FIG. **3**), a (2-2).sup.th side portion **322** (e.g., the (2-1).sup.th sidewall **221***a* in FIG. **3**) extending from the (2-1).sup.th side portion **321** and disposed in a direction (e.g., the vertical direction) different from that of the (2-1).sup.th side portion **321**, and a (2-3).sup.th side portion **323** (e.g., the (2-3).sup.th sidewall **221***c* in FIG. **3**) extending from the (2-2).sup.th side portion **322** and disposed in the same direction as the (2-1).sup.th side portion **321**.
[0102] According to various embodiments, the first frame **310** and the second frame **320** may provide an outer surface of the electronic device **101** and may be disposed to be spaced apart from each other. For example, a segmented portion **461** may be provided between the first frame **310** and the second frame **320** to be electrically separated from each other. As another example, the first frame **310** may be designed to have a "]" shape, and the second frame **320** may be designed to have a "[" shape. In the state in which the first housing **301** is closed relative to the second housing **302**, the first frame **310** and the second frame **320** may be designed to have a substantially rectangular frame shape (including the segmented portion **461**). In the state in which the first housing **301** is opened relative to the second housing **302**, the segmented portion **461** may be expanded.
[0103] According to various embodiments, the segmented portion **461** may have various shapes (e.g., materials) in which the first frame **310** and the second frame **320** may be physically/electrically spaced apart from each other. For example, the segmented portion **461** may be formed as an air gap to separate the first frame **310** and the second frame **320** from each other. As another example, the segmented portion **461** may be referred to as an opening, a recess, or a groove.
[0104] According to various embodiments, in the closed state, the non-conductive portion **462** overlapping the antenna radiator (e.g., the second conductive portion **420**) of the second frame **320** may be formed of an insulating material to electrically separate the metal portion of the first frame

310 and the antenna radiator portion of the second frame **320**. The non-conductive portion **462** may provide a dielectric constant different from that of a conductive portion (e.g., the first conductive portion **410** and/or the second conductive portion **420**). For example, the non-conductive portion **462** may include any material that is insulative, including an elastomer material, ceramic, mica, glass, plastic, a metal oxide, air, and/or other materials that are superior in insulation to metal, but not limited to these materials.

[0105] According to various embodiments, at least a portion of the first frame **310** or at least a portion of the second frame **320** may be disposed not to overlap each other.

[0106] For example, when a portion of the second frame **320** (e.g., the (2-3).sup.th side portion **323**) is used as an antenna radiator, the portion may be disposed always (e.g., when the first frame **310** relative to the second frame **320** are in the closed and opened states) not to overlap a portion of the first frame **310** (e.g., the (1-3).sup.th side portion **313**). As another example, when a portion of the second frame **320** (e.g., the (2-1).sup.th side portion **321**) is used as an antenna radiator, the portion may be disposed always (e.g., when the first frame **310** and the second frame **320** are in the closed and opened states) not to overlap a portion of the first frame **310** (e.g., the (1-1).sup.th side portion **311**).

[0107] According to various embodiments, in addition to the segmented portion (e.g., the segmented portion **461**) provided between the first frame **310** and the second frame **320**, each frame may include a plurality of segmented portions. For example, as illustrated in FIG. **6**, in the second frame **320**, one segmented portion **461***a* segmenting the (2-1).sup.th side portion **321**, two segmented portions **461***b* and **461***c* segmenting the (2-2).sup.th side portion **322**, and one segmented portion **461***d* segmenting the (2-3).sup.th side portion **323**. However, the plurality of segmented portions correspond to an example, and the segmented portions may be changed in various ways according to a frequency range used by an antenna.

[0108] According to various embodiments, by providing the first frame **310** and the second frame **320** such that unlike a metal antenna structure of an existing slidable terminal, a structure in which a portion of the first frame **310** used as an antenna radiator and the second frame **320** overlap each other can be avoided and a separation operation can be generated with reference to the segmented portion, it is possible to reduce a deviation in antenna performance physically generated in the state in which the screen of the display **203** is contracted or expanded.

[0109] FIG. **7**A is an enlarged perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is closed according to various embodiments.

[0110] FIG. **7**B is an enlarged perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is open according to various embodiments.

[0111] According to various embodiments, the electronic device **101** may include a first housing (e.g., the first housing **301** in FIGS. **5**A and **5**B) and a second housing (e.g., the second housing **302** in FIGS. **5**A and **5**B). The electronic device **101** may further include an antenna structure, wherein at least one component (e.g., a conductive portion) of the antenna structure may include a portion of the first housing **301** and/or the second housing **302**.

[0112] According to various embodiments, as the first housing **301** (and the flexible display (not illustrated) connected to the first housing **301**) slides in or out relative to the second housing **302**, the electronic device **101** may be in the closed state or the opened state.

[0113] According to various embodiments, the first housing **301** may include a first conductive portion **410**, a first segmented portion **431** extending from the first conductive portion **410**, and a first non-conductive portion **432** disposed adjacent to the first segmented portion **431**.

[0114] According to various embodiments, the second housing **302** may include a second conductive portion **420**, and the second conductive portion **420** may be used as a radiator of an antenna structure. According to an embodiment, at least a portion of the second housing **302** may

include a metal frame (e.g., the second frame **320** in FIGS. **5**A to **6**), and at least a portion of the metal frame may be the second conductive portion **420**.
[0115] According to an embodiment, the second conductive portion **420** may include a recess **422** in which at least a portion of the first non-conductive portion **432** is slidable. For example, the recess **422** may have a groove shape formed along the inner surface of the second conductive portion **420**. The recess **422** may be designed to have a size corresponding to at least a portion of the non-conductive portion **432**, and may guide the sliding movement of the non-conductive portion **432**.
[0116] According to various embodiments, the first segmented portion **431** may include an insulating material to separate the first conductive portion **410** and the second conductive portion **420** from each other, and thus the first conductive portion **410** or the second conductive portion **420** may operate as an antenna radiator. According to an embodiment, the first segmented portion **431** and the first non-conductive portion **432** may be integrally configured. The first segmented portion **431** and the first non-conductive portion **432** may be designed to have a stepped shape. For example, the first segmented portion **431** may be a portion extending from one end of the first conductive portion **410** in the −Y-axis direction, and the first non-conductive portion **432** may be a portion extending from the first conductive portion **410** toward the second conductive portion **420** to be movable along the recess **422**.
[0117] According to an embodiment, the first segmented portion **431** may be located between the first conductive portion **410** and the second conductive portion **420** in order to substantially prevent and/or reduce the first conductive portion **410** and the second conductive portion **420** from coming into contact with each other. As illustrated in **7**A and **7**B, when viewed from the front side of the first housing **301** (and the second housing **302**) (e.g., in the −Z-axis direction), the (1-1).sup.th portion **431** may be configured to extend in a second direction (e.g., the Y-axis direction) perpendicular to a first direction {circle around (**1**)} (e.g., the slide movement direction). For example, the top surface **431***a* (e.g., the surface oriented in the Y-axis direction) of the first segmented portion **431** may be located on substantially the same plane as the top surface **411** of the first conductive portion **410** and/or the top surface **421** of the second conductive portion **420**. The bottom surface **431***b* (e.g., the surface oriented in the −Y-axis direction) of the first segmented portion **431** may be located in a lower direction (e.g., the −Y-axis direction) than the bottom surface **412** of the first conductive portion **410**.
[0118] According to an embodiment, at least a portion of the first non-conductive portion **432** may slide along the recess **422** of the second conductive portion **420** in the first direction {circle around (**1**)}. As illustrated in FIGS. **7**A and **7**B, when viewed from the front side of the first housing **301** (and the second housing **302**) (e.g., in the −Z-axis direction), the first non-conductive portion **432** may be configured to extend from the lower end of the first segmented portion **431** in a direction opposite to the first direction {circle around (**1**)} (e.g., the slide movement direction).
[0119] According to various embodiments, the first conductive portion **410** and the second conductive portion **420** may be always spaced apart from each other. According to an embodiment, when viewing the first housing **301** and the second housing **302** in the Y-axis direction, the first conductive portion **410** and the second conductive portion **420** may not overlap each other. For example, from the closed state (e.g., FIG. **7**A) to the opened state (e.g., FIG. **7**B) of the first housing **301** relative to the second housing **302**, the first conductive portion **410** and the second conductive portion **420** may maintain a spaced-apart state.
[0120] According to an embodiment, in the closed state of the first housing **301** relative to the second housing **302** (e.g., FIG. **7**A), the first non-conductive portion **432** overlaps the second conductive portion **420**. In the open state of the first housing **301** relative to the second housing **302** (e.g., FIG. **7**B), at least a portion of the first non-conductive portion **432** may be disposed to overlap the second conductive portion **420**.
[0121] According to an embodiment, in the closed state, the first conductive portion **410** may be

disposed to be spaced apart from the second conductive portion **420** with the first segmented portion **431** interposed therebetween. In the closed state, the top surface **411** of the first conductive portion **410**, the top surface **431***a* of the first segment **431**, and the top surface **421** of the second conductive portion **420** are exposed to the outside, and may form substantially one plane. The first non-conductive portion **432** may be located entirely within the recess **422** of the second conductive portion **420**.

[0122] According to an embodiment, in the opened state, the first conductive portion **410** may be disposed to be spaced apart from the second conductive portion **420** with the first segmented portion **431** interposed therebetween. In the opened state, due to the sliding movement, the top surface **411** of the first conductive portion **410** and the top surface **431***a* of the first segmented portion **431** may form a relatively increased separation distance from the top surface **421** of the second conductive portion **420** compared with the closed state. A portion of the first non-conductive portion **432** may be exposed to the outside as it slides, and the other portion may be located within the recess **422** of the second conductive portion **420**.

[0123] According to an embodiment, since the first conductive portion **410** and the second conductive portion **420** are spaced apart from each other in the closed state and the open state, it is possible to reduce a deviation in antenna performance physically generated in the contracted or expanded state of the flexible display screen when at least a portion of the second conductive portion **420** is used as an antenna radiator. In addition, it is possible to provide an antenna structure in which resonance characteristics can be maintained similarly between the closed operation and the opened operation.

[0124] FIG. **8**A is a diagram illustrating an internal antenna structure of the electronic device in the state in which the electronic device is closed according to various embodiments.

[0125] FIG. **8**B is a diagram illustrating an internal antenna structure of the electronic device in the state in which the electronic device is open according to various embodiments.

[0126] According to various embodiments, the electronic device **101** may include a first housing **301**, a second housing **302**, a main circuit board **204**, a first sub-board **500***a*, and/or a second sub-board **500***b*.

[0127] The configurations of the first housing **301** and the second housing **302** of FIGS. **8**A and **8**B may be wholly or partly the same as or similar to those of the first housing **301** and the second housing **302** of FIGS. **5**A to **6**. The configuration of the main circuit board **204** of FIGS. **8**A and **8**B may be wholly or partly the same as or similar to that of the circuit board **204** of FIG. **4**.

[0128] According to various embodiments, the main circuit board **204** and the second sub-board **500***b* may be disposed in the second housing **302**. The main circuit board **204** and the second sub-board **500***b* may be electrically connected to each other by a connection member **523** (e.g., an RF cable).

[0129] According to an embodiment, a processor, a memory, and/or an interface may be disposed on the main circuit board **204**. The processor may include, for example, one or more of a central processing unit, an application processor, a graphics processing unit, an image signal processor, a sensor hub processor, or a communication processor. According to various embodiments, the main circuit board **204** may include a flexible printed circuit board type radio frequency cable (RFC). For example, the main circuit board **204** may be electrically connected to an antenna structure (e.g., the antenna module **197** in FIG. **1**) and a communication module (e.g., the communication module **190** in FIG. **1**).

[0130] According to an embodiment, the memory may include, for example, a volatile memory or a nonvolatile memory. According to an embodiment, the interface may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, an SD card interface, and/or an audio interface. The interface may electrically or physically connect, for example, the electronic device **101** to an external electronic device and may include a USB connector, an SD card/an MMC connector, or an audio connector.

[0131] According to various embodiments, the first sub-board **500***a* may be disposed in the first housing **301**. The main circuit board **204** and the first sub-board **500***a* may be electrically connected to each other by a flexible connection member **513** (e.g., an RF cable).
[0132] According to various embodiments, the first antenna structure **400***a* may be located in the first housing **301**. The first antenna structure **400***a* may include a first antenna pattern (e.g., the first conductive portion **410**), a first feeding portion **511**, and a first ground portion **512**. The first housing **301** may include a first conductive portion **410** formed of a metal frame, and at least a portion of the first conductive portion **410** may be used as the first antenna pattern (e.g., an antenna radiator). For example, the first conductive portion **511** acting as an antenna may be configured to transmit and/or receive radio frequency (RF) signals.
[0133] According to various embodiments, at one end (or both ends) of the first conductive portion **410**, a non-conductive portion (e.g., the first segmented portion **431**) is provided, and the first conductive portion **410** and the first segmented portion **431** may be designed to have an arbitrary length in order to implement an antenna of a desired frequency band. Although not illustrated in the drawings, the first feeding portion **511** and/or the first ground portion **512** may include a contact structure (e.g., a C-Clip or the like) for electrically connecting to the first sub-board **500***a*.
[0134] According to various embodiments, the first feeding portion **511** may be a portion extending from the first conductive portion **410** of the first housing **301** to the inside of the electronic device **101**. The first feeding portion **511** may be electrically connected to the first sub-board **500***a* disposed in the first housing **301** to transmit current to the first antenna pattern (e.g., the first conductive portion **410**). According to an embodiment, the first sub-board **500***a* may be electrically connected to the main circuit board **204** by the flexible connection member **513** (e.g., an RF cable). For example, when the first housing **301** is slid relative to the second housing **302** from the closed state to the opened state, the flexible connection member **513**, which is designed as a flexible cable, may be deformable to accommodate an increase in the separation distance of the first housing **301** to the first housing **301**. Accordingly, from the closed state to the opened state, the electrical connection to the main circuit board **204** for using the first conductive portion **410** as an antenna may be always maintained.
[0135] According to an embodiment, a transmission/reception (Tx/Rx) terminal of the communication circuit disposed on the main circuit board **340** may be connected to the first feeding portion **511** to perform communication. The electronic device **101** may include any type of elements for impedance matching, such as a switch, a resistive element, a capacitive element, an inductive element, or any combination thereof. The elements may be used to selectively change the antenna frequency band. In the illustrated embodiment, the first feeding portion **511** extends in a direction perpendicular to the longitudinal direction of the first conductive portion **410**, but is not limited thereto. The first feeding portion may be design-changed to have another configuration having a different thickness and/or angle.
[0136] According to various embodiments, the first ground portion **512** may be disposed between the first segmented portion **431** and the first feeding portion **511**. For example, the first ground portion **512** may be electrically connected to a portion of the first conductive portion **410**. According to an embodiment, the first ground portion **512** may include an element or a switch for changing an impedance for adjusting a frequency band. For example, the first ground portion **512** may include an element that provides various electrical lengths by switching to a direct ground without passing through an inductor, a capacitor, a combination thereof, or an intermediate element at a ground stage to create resonances of various lengths in the physically identical first antenna pattern. As another example, the first ground portion **512** may be at least a portion extending from the first conductive portion **410** to the inside of the first housing **301**. According to an embodiment, the first ground portion **512** may be connected to a portion of the first sub-board **500***a* in the first housing **301** directly or via a conductive member (not illustrated). For example, the first ground portion **512** may be electrically connected to a ground of the first sub-board **500***a* in the first

housing **301**. For example, the conductive member may include at least one of members such as a wire, a clip (e.g., a c-clip), a screw, and a conductive sponge.
[0137] According to an embodiment, the first ground portion **512** may be a ground of an element electrically connected to a feeding portion of another antenna (e.g., another conductive portion).
[0138] According to various embodiments, the second antenna structure **400***b* may be located in the second housing **302**. The second antenna structure **400***b* may include a second antenna pattern (e.g., the second conductive portion **420**), a second feeding portion **521**, and a second ground portion **522**. The second housing **302** may include a second conductive portion **420** formed of a metal frame, and at least a portion of the second conductive portion **420** may be used as the second antenna pattern (e.g., an antenna radiator). For example, the second conductive portion **420** acting as an antenna may be configured to transmit and/or receive radio frequency (RF) signals. The configurations of the first conductive portion **410**, the first feeding portion **511**, and the first ground portion **512** of the first antenna structure **400***a* may be applicable to the configurations of the second conductive portion **420**, the second feeding portion **521**, and the second ground portion **522** of the second antenna structure **400***b*.
[0139] FIG. **9** is a graph showing antenna performance related to the first antenna structure by the sliding operation between the housings in FIGS. **8**A and **8**B according to various embodiments.
[0140] FIG. **10** is a graph showing antenna performance related to the second antenna structure by the sliding operation between the housings in FIGS. **8**A and **8**B according to various embodiments.
[0141] Referring to FIGS. **9** and **10**, when the state in which the second housing is closed relative to the first housing (e.g., FIG. **8**A) is changed to the state in which the first housing is opened relative to the second housing (e.g., FIG. **8**B), the separation distance (g) between the second housing and the first housing was measured to vary (increase) from 1 mm to 5 mm (e.g., g=1, g=2, g=3, g=4, and g=5). For example, when the closed state, the first conductive portion (e.g., the first conductive portion **410** in FIGS. **7**A to **8**B) and the second conductive portion (e.g., the second conductive portion **420** in FIGS. **7**A to **8**B) were designed such that the distance between the first and second conductive portions (e.g., the first segmented portion **431** in FIGS. **7**A to **8**B)) is about 1 mm, and modeling was performed such that, when a slide operation is performed, the separation distance between the first conductive portion and the second conductive portion is gradually increased. When the separation distance was greater than a certain distance, the effect between the first conductive portion and the second conductive portion was insignificant. Thus, the maximum separation distance was simulated up to approximately 5 mm. The change in antenna performance was confirmed using S-parameters.
[0142] Referring to FIG. **9**, it can be seen that the frequency of the antenna using the first conductive portion in FIGS. **8**A and **8**B is about 2100 MHz. When the electronic device including the first conductive portion, the second conductive portion, and the first segmented portion are provided according to the disclosure, it can be seen that there is little influence on the antenna performance according to the sliding operation of the first housing relative to the second housing. With the electronic device according to the disclosure, it is possible to provide an antenna structure in which resonance characteristics may be similarly maintained between slide-in and slide-out operations.
[0143] Referring to FIG. **10**, it can be seen that the frequency of the antenna using the second conductive portion in FIGS. **8**A and **8**B is about 900 MHz. When the first conductive portion, the second conductive portion, and the first segmented portion are provided according to the disclosure, it can be seen that there is little influence on the antenna performance according to the sliding operation of the first housing relative to the second housing. With the electronic device according to the disclosure, it is possible to provide an antenna structure in which resonance characteristics may be similarly maintained between slide-in and slide-out operations.
[0144] FIG. **11**A is an enlarged perspective view of an area of the first housing and the second housing for illustrating a segmented portion in the state in which the electronic device is closed

according to various embodiments.

[0145] FIG. **11**B is an enlarged perspective view of an area of the first housing and the second housing for illustrating a segmented portion in the state in which the electronic device is open according to various embodiments.

[0146] According to various embodiments, the electronic device **101** may include a first housing (e.g., the first housing **301** in FIGS. **5**A and **5**B) and a second housing (e.g., the second housing **302** in FIGS. **5**A and **5**B). The electronic device **101** may further include an antenna structure, wherein at least one component (e.g., the first and second conductive portions) of the antenna structure may be an area of the first housing **301** and/or the second housing **302**.

[0147] The configurations of the first conductive portion **410** and the second conductive portion **420** of FIGS. **11**A and **11**B may be partly or wholly the same as or similar to the configurations of the first conductive portion **410** and the second conductive portion **420** of FIGS. **5**A to **7**B.

[0148] According to various embodiments, as the first housing **301** (and the flexible display (not illustrated) connected to the first housing **301**) slides in or out relative to the second housing **302**, the electronic device may be in the closed state or the opened state. Hereinafter, a structure different from that of FIGS. **7**A and **7**B will be mainly described.

[0149] According to various embodiments, the first housing **301** may include a first conductive portion **410**, a first segmented portion **431** extending from the first conductive portion **410**, and a (1-1).sup.th non-conductive portion **432** disposed adjacent to the first segmented portion **431**.

[0150] According to various embodiments, the second housing **302** may include a second conductive portion **420** and a second segmented portion **440** extending from the second conductive portion **420**, and the second conductive portion **420** may be used as a radiator of an antenna structure. According to an embodiment, the second conductive portion **420** and the second segmented portion **440** may include a recess **422** in which at least a portion of the (1-1).sup.th non-conductive portion **432***a* is slidable. The recess **422** may be designed to have a size corresponding to, for example, at least a portion of the (1-1).sup.th non-conductive portion **432***a* to guide the slide movement of the (1-1).sup.th non-conductive portion **432***a*.

[0151] According to various embodiments, the second segmented portion **440** may include an insulating material, and may be disposed at opposite ends of the second conductive portion **420** together with the first segmented portion **431**. The length of the second conductive portion **420** may be arbitrarily adjusted according to the position of the second segmented portion **440**, and accordingly, the frequency band of the antenna may be selectively adjusted. As illustrated, the second segmented portion **440** is formed to segment the upper frame (e.g., the (2-1).sup.th side portion **321** in FIG. **5**B) of the second housing **302**, but is not limited thereto. The second segmented portion **440** may be design-changed to segment an area of the left frame (e.g., the (2-2).sup.th side portion **322** in FIG. **5**B) or the lower frame (e.g., the (2-3).sup.th side portion **323** in FIG. **5**B) of the second housing **302**.

[0152] According to various embodiments, the first segmented portion **431** and the (1-1).sup.th non-conductive portion **432***a* may be designed to have a stepped shape. For example, the first segmented portion **431** may extend from one end of the first conductive portion **410**, and the (1-1).sup.th non-conductive portion **432***a* may extend from the first segmented portion **431** to be movable along the recess **422**.

[0153] According to an embodiment, the first segmented portion **431** may be located between the first conductive portion **410** and the second conductive portion **420** in order to substantially prevent and/or reduce the first conductive portion **410** and the second conductive portion **420** from coming into contact with each other. According to an embodiment, the (1-1).sup.th non-conductive portion **432***a* in FIG. **11**B may be designed to have a longer length than the first non-conductive portion **432** of FIG. **7**B. The length of the (1-1).sup.th non-conductive portion **432***a* may be determined depending on a moving distance of the first housing **301**. For example, since the (1-1).sup.th non-conductive portion **432***a*, which is formed of an insulator, does not substantially affect the antenna radiation performance, the (1-1).sup.th non-conductive portion may be designed to extend to the second segmented portion **440**. However, the length of the (1-1).sup.th non-conductive portion **432***a* is not limited to extending beyond the second segmented portion **440** as illustrated in the drawings, and may be design-changed to have various lengths that do not reach the second segmented portion **440**.

[0154] FIG. **12**A is an enlarged perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is closed according to various embodiments.

[0155] FIG. **12**B is an enlarged perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is open according to various embodiments.

[0156] According to various embodiments, the electronic device **101** may include a first housing (e.g., the first housing **301** in FIGS. **5**A and **5**B) and a second housing (e.g., the second housing **302** in FIGS. **5**A and **5**B). The electronic device **101** may further include an antenna structure, wherein at least one component (e.g., the first and second conductive portions) of the antenna structure may be an area of the first housing **301** and/or the second housing **302**.

[0157] The configurations of the first conductive portion **410** and the second conductive portion **420** of FIGS. **12**A and **12**B may be partly or wholly the same as or similar to the configurations of the first conductive portion **410** and the second conductive portion **420** of FIGS. **5**A to **7**B.

[0158] According to various embodiments, as the first housing **301** (and the flexible display (not illustrated) connected to the first housing **301**) slides in or out relative to the second housing **302**, the electronic device may be in the closed state or the opened state. Hereinafter, a structure different from that of FIGS. **7**A and **7**B will be mainly described.

[0159] According to various embodiments, the first housing **301** may include a first conductive portion **410**, a (1-1).sup.th segmented portion **430***a* extending from the first conductive portion **410**, and a (1-2).sup.th non-conductive portion **432***b* extending from the (1-1).sup.th segmented portion **430***a*. For example, the (1-1).sup.th segmented portion **430***a* and the (1-2).sup.th non-conductive portion **432***b* may be formed of the same material in an integrated form.

[0160] According to various embodiments, the second housing **302** may include a second conductive portion **420**, and the second conductive portion **420** may be used as a radiator of an antenna structure. According to an embodiment, the second conductive portion **420** may include a recess **422** in which at least a portion of the (1-2).sup.th non-conductive portion **432** is slidable. The recess **422** may be designed to have a size corresponding to at least a portion of the (1-2).sup.th non-conductive portion **432***b* to guide the sliding movement of the (1-2).sup.th non-conductive portion **432***b*.

[0161] According to various embodiments, the (1-1).sup.th segmented portion **430***a* and the (1-2).sup.th non-conductive portion **432***b* may include an insulating material and may have a thickness corresponding to (e.g., the same as) the first conductive portion **410**. In the closed state, the (1-2).sup.th non-conductive portion **432***b* is accommodated in the recess **422**, but an area adjacent to the first conductive portion **410** (e.g., the (1-1).sup.th segmented portion **430***a*) may be exposed to the outside to space the first conductive portion **410** and the second conductive portion **420** apart from each other.

[0162] According to various embodiments, the top surface **431***aa* (e.g., the surface oriented in the +Y-axis direction) of the (1-1).sup.th segmented portion **430***a* may be located on the same plane as the top surface **411** of the first conductive portion **410**, but may not be located on the same plane as the top surface **421** of the second conductive portion **420**. For example, the top surface **421** of the second conductive portion **420** protrudes upward in the +Y-axis direction with respect to the top surface **431***aa* of the (1-1).sup.th segmented portion **430***a* and the top surface **411** of the first conductive portion **410**.

[0163] According to various embodiments, in the closed state and the opened state, the second conductive portion **420** to be used as an antenna radiator is spaced apart from the first conductive portion **410** by the (1-2).sup.th non-conductive portion **432***b*. Thus, it is possible to reduce a deviation in antenna performance physically occurring in the state in which the flexible display screen is contracted or expanded.

[0164] FIG. **13**A is a perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is closed according to various embodiments.

[0165] FIG. **13**B is a perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is open according to various embodiments of the disclosure.

[0166] According to various embodiments, the electronic device **101** may include a first housing **301**, a second housing **302**, and a flexible display **303**. The electronic device **101** may further include an antenna structure, wherein at least one component (e.g., a conductive portion) of the antenna structure may be located in an area of the first housing **301** and/or the second housing **302**.

[0167] The configurations of the first housing **301** and the second housing **302** of FIGS. **13**A and **13**B may be partly or wholly the same as or similar to those of the first housing **301** and the second housing **302** of FIGS. **5**A to **7**B.

[0168] According to various embodiments, as the first housing **301** (and the flexible display **303** connected to the first housing **301**) slides in or out relative to the second housing **302**, the electronic device may be in the closed state or the opened state. Hereinafter, a structure different from that of FIGS. **7**A and **7**B will be mainly described.

[0169] According to various embodiments, the first housing **301** may include a first conductive portion **410** and a (1-3).sup.th segmented portion **430***c* extending from the first conductive portion **410**, and the first conductive portion **410** may be used as a radiator of an antenna structure. According to an embodiment, the first conductive portion **410** may be a radiator having a relatively larger length compared to the second conductive portion **420**.

[0170] According to various embodiments, the second housing **302** may include a second conductive portion **420** and a second segmented portion **470**, and the second conductive portion **420** may be used as a radiator of an antenna structure. According to an embodiment, the second conductive portion **420** may include a recess (not illustrated) in which at least a portion of the (1-3).sup.th segmented portion **430***c* is slidable. The recess may be designed to have a size corresponding to at least a portion of the (1-3).sup.th segmented portion **430***c* to guide the slide movement of the (1-3).sup.th segmented portion **430***c*.

[0171] FIG. **14**A is a perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is closed according to various embodiments.

[0172] FIG. **14**B is a perspective view of an area of the first housing and the second housing illustrating a segmented portion in the state in which the electronic device is open according to various embodiments.

[0173] According to various embodiments, the electronic device **101** may include a first housing **301**, a second housing **302**, and a flexible display **303**. The electronic device **101** may further include an antenna structure, wherein at least one component (e.g., a conductive portion) of the antenna structure may be an area of the first housing **301** and/or the second housing **302**.

[0174] The configurations of the first housing **301** and the second housing **302** of FIGS. **14**A and **14**B may be partly or wholly the same as or similar to those of the first housing **301** and the second housing **302** of FIGS. **5**A to **7**B.

[0175] According to various embodiments, as the first housing **301** (and the flexible display **303** connected to the first housing **301**) slides in or out relative to the second housing **302**, the electronic device may be in the closed state or the opened state. Hereinafter, a structure different from that of FIGS. **7**A and **7**B will be mainly described.

[0176] According to various embodiments, the first housing **301** may include a first conductive portion **410** and a (1-4).sup.th segmented portion **430***d* extending from the first conductive portion **410**, and the first conductive portion **410** may be used as a radiator of an antenna structure. According to an embodiment, the first conductive portion **410** may be a radiator having a relatively smaller length compared to the second conductive portion **420**. The (1-4).sup.th segmented portion **430***d* may be designed to have a larger length compared to the (1-3).sup.th segmented portion **430***c* in FIGS. **13**A and **13**B to overlap most of the second conductive portion **420**.
[0177] According to various embodiments, the second housing **302** may include a second conductive portion **420** and a second segmented portion **470**, and the second conductive portion **420** may be used as a radiator of an antenna structure. According to an embodiment, the second conductive portion **420** may include a recess (not illustrated) in which at least a portion of the (1-4).sup.th segmented portion **430***d* is slidable. The recess may be designed to have a size corresponding to at least a portion of the (1-4).sup.th segmented portion **430***d* to guide the slide movement of the (1-4).sup.th segmented portion **430***d*.
[0178] An electronic device (e.g., the electronic device **101** in FIGS. **1** to **4**) according to various embodiments of the disclosure may include: a first housing (e.g., **201** in FIG. **4**) including a first conductive portion (e.g., **410** in FIG. **7**A), a first non-conductive portion (e.g., **431** in FIG. **7**A), and a first segmented portion (e.g., **432** in FIG. **7**A) extending from the first conductive portion, a second housing (e.g., **202** in FIG. **4**) configured to accommodate at least a portion of the first housing and to guide a slide movement of the first housing, the second housing including a second conductive portion (e.g., **420** in FIG. **7**A), and a flexible display (e.g., **203** in FIG. **4**) including a first area connected to the first housing and a second area extending from the first area and configured to be bendable or rollable. From a slide-in state to a slide-out state of the first housing with respect to the second housing, the first conductive portion and the second conductive portion may be spaced apart from each other. In the slide-in state of the first housing with respect to the second housing, at least a portion of the first non-conductive portion may overlap the second conductive portion.
[0179] According to various embodiments, the first conductive portion and the second conductive portion may be spaced apart from each other with the first segmented portion interposed therebetween.
[0180] According to various embodiments, the first segmented portion may include an air gap or an insulating material.
[0181] According to various embodiments, in the slide-out state of the first housing with respect to the second housing, at least a portion of the first non-conductive portion may overlap the second conductive portion.
[0182] According to various embodiments, the second conductive portion may include a recess (e.g., **422** in FIG. **7**A) formed along an inner surface of the second conductive portion configured to guide the slide movement of the first non-conductive portion.
[0183] According to various embodiments, an end surface of the first conductive portion and an end surface of the second conductive portion may face each other with the first segmented portion interposed therebetween.
[0184] According to various embodiments, one surface (e.g., **411** in FIG. **7**A) of the first conductive portion exposed to the outside and one surface (e.g., **421** in FIG. **7**A) of the second conductive portion exposed to the outside may be located on the same plane.
[0185] According to various embodiments, the second conductive portion may be configured to be operable as an antenna radiator.
[0186] According to various embodiments, the first segmented portion and the non-conductive portion may be formed a stepped shape.
[0187] According to various embodiments, the first segmented portion may extend from one end of the first conductive portion in a direction perpendicular to the slide direction, and the non-

conductive portion may extend from the first segmented portion in the slide direction to be movable along the recess of the second conductive portion.

[0188] According to various embodiments, a top surface (e.g., **431***a* in FIG. **7**B) of the first segmented portion may be located on the same plane as a top surface of the first conductive portion and/or a top surface of the second conductive portion, and a bottom surface of the first segmented portion (e.g., **431***b* in FIG. **7**B) may be configured to further extend in a lower direction than the bottom surface of the first conductive portion.

[0189] According to various embodiments, one end of the second conductive portion may face the first segmented portion, a second segmented portion (e.g., **440** in FIG. **11**B) may be disposed on an other end of the second conductive portion, and the second conductive portion and the second segmented portion may include a recess formed along inner surfaces thereof to guide the slide movement of the first non-conductive portion.

[0190] According to various embodiments, the first segmented portion and the non-conductive portion may include an insulating material and may have an extending shape having a thickness corresponding to the first conductive portion.

[0191] According to various embodiments, a top surface of the second conductive portion may extend further in an upward direction with respect to the top surface of the first segmented portion and the top surface of the first conductive portion.

[0192] According to various embodiments, in the slide-in state of the first housing with respect to the second housing, a resonance characteristic formed in the second conductive portion may correspond to a resonance characteristic formed in the second conductive portion in the slide-out state of the first housing with respect to the second housing.

[0193] According to various embodiments, the electronic device may further include a main circuit board (e.g., **204** in FIG. **8**A) disposed in the second housing, a first sub-board (e.g., **500***a* in FIG. **8**A) disposed in the first housing, and a flexible connection member (e.g., **513** in FIG. **8**A) configured to be deformable to electrically connect the main circuit board and the first sub-board from the slide-in state to the slide-out state.

[0194] An electronic device (e.g., the electronic device **101** in FIGS. **1** to **4**) according to various embodiments of the disclosure may include: a first housing (e.g., **201** in FIG. **4**) including a first conductive portion (e.g., **410** in FIG. **7**A), a first non-conductive portion (e.g., **432** in FIG. **7**A), and a first segmented portion (e.g., **431** in FIG. **7**A) extending from the first conductive portion, a second housing (e.g., **202** in FIG. **4**) configured to receive at least a portion of the first housing, guide a slide movement of the first housing, and including a second conductive portion (e.g., **420** in FIG. **7**A), and a flexible display (e.g., **203** in FIG. **4**) including a first area connected to the first housing and a second area extending from the first area and configured to be bendable or rollable. From a slide-in state to a slide-out state of the first housing with respect to the second housing, the first conductive portion and the second conductive portion may be spaced apart from each other with the first segmented portion interposed therebetween.

[0195] According to various embodiments, in the slide-in state of the first housing with respect to the second housing, at least a portion of the first non-conductive portion may overlap the second conductive portion.

[0196] According to various embodiment, in the slide-out state of the first housing with respect to the second housing, at least a portion of the first non-conductive portion may overlap the second conductive portion.

[0197] According to various embodiment, the second conductive portion may include a recess formed along an inner surface thereof configured to guide the slide movement of the first non-conductive portion.

[0198] According to various embodiments, an end surface of the first conductive portion and an end surface of the second conductive portion may face each other with the first segmented portion interposed therebetween.

[0199] While the disclosure has been illustrated and described with reference to various embodiments, it will be understood that the various embodiments are intended to be illustrative, not limiting. It will be further understood by those skilled in the art that various changes, substitutions and modifications may be made without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

Claims

1. An electronic device comprising: a flexible display; a first housing configured to accommodate a portion of the flexible display, the first housing including a first conductive portion forming a portion of an external side of the electronic device; and a second housing configured to accommodate another portion of the flexible display and configured to slide in and slide out with respect to the first housing in a lengthwise direction of the external side, the second housing including a second conductive portion forming a portion of the external side surface, wherein, when the second housing is slid in with respect to the first housing, a first surface of the first conductive portion configured to form an external side of the electronic device is to be substantially coplanar with a first surface of the second conductive portion configured to form an external side of the electronic device.

2. The electronic device of claim 1, wherein, when the second housing is slid in with respect to the first housing, a second surface of the first conductive portion which is substantially perpendicular to the first surface of the first conductive portion is to substantially face a second surface of the second conductive portion which is substantially perpendicular to the first surface of the second conductive portion.

3. The electronic device of claim 2, wherein the second housing includes a first non-conductive portion disposed between the second surface of the first conductive portion and the second surface of the second conductive portion.

4. The electronic device of claim 3, wherein the first non-conductive portion is formed such that, when the second housing is slid in with respect to the first housing, an external surface of the first non-conductive portion configured to form an external side of the electronic device is to be substantially coplanar with the first surface of the first conductive portion and the first surface of the second conductive portion.

5. The electronic device of claim 3, wherein the second housing includes a second non-conductive portion formed as protruding from the first non-conductive portion in the length direction such that, when the second housing is slid in with respect to the first housing, at least part of the second non-conductive portion is to be slide into a recess formed in the first conductive portion.

6. The electronic device of claim 5, wherein the second non-conductive portion is formed such that, when the second housing is slid out with respect to the first housing, at least part of the second non-conductive portion is to be slide out of the recess.

7. The electronic device of claim 1, wherein the first conductive portion is configured to operate as an antenna radiator.

8. The electronic device of claim 1, wherein the second housing includes a third conductive portion forming a portion of the external side of the electronic device and spaced apart from the second conductive portion at least by a non-conductive portion.

9. The electronic device of claim 8, wherein the third conductive portion is configured to operate as an antenna radiator.

10. The electronic device of claim 1, wherein, when the second housing is slid out with respect to the first housing, the first conductive portion and the second conductive portion are spaced apart from each other.

11. The electronic device of claim 3, wherein the first non-conductive portion includes an air gap or an insulating material.
12. The electronic device of claim 4, wherein, when the second housing is slid in with respect to the first housing, a first surface of the first conductive portion overlaps the external surface of the first non-conductive portion, when viewed in direction perpendicular to the lengthwise direction.
13. The electronic device of claim 4, wherein, when the second housing is slid out with respect to the first housing, at least a portion of a first surface of the first conductive portion overlaps the external surface of the first non-conductive portion, when viewed in direction perpendicular to the lengthwise direction.
14. The electronic device of claim 3, wherein first non-conductive portion and the second conductive portion are formed a stepped shape.
15. The electronic device of claim 5, wherein the first non-conductive portion and the second non-conductive portion include an insulating material.
16. The electronic device of claim 5, wherein the first non-conductive portion and the second non-conductive portion have an extending shape having a thickness corresponding to the second conductive portion.
17. The electronic device of claim 1, wherein, in the slide-in state of the second housing with respect to the first housing, a resonance characteristic in the first conductive portion corresponds to a resonance characteristic formed in the first conductive portion in the slide-out state of the second housing with respect to the first housing.
18. The electronic device of claim 1, further comprising: a main circuit board disposed in the first housing; a first sub-board disposed in the second housing; and a flexible connection member configured to be deformable to electrically connect the main circuit board and the first sub-board from the slide-in state to the slide-out state.
19. The electronic device of claim 1, wherein one end of the second conductive portion segmented by a segmented portion.
20. The electronic device of claim 19. wherein a segmented portion disposed between the first conductive portion and the second conductive portion.